

Surface energies and Wulff equilibrium shapes of Cu and Pd nanoparticles

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This report presents a density functional theory (DFT) study of the surface energies of the three low-index facets — (001) , (011) , and (111) — of copper and palladium, two face-centred cubic (fcc) metals with contrasting catalytic properties. Both fixed-geometry and fully relaxed surface energies are computed from symmetric slab models using the expression $\gamma = (E_{\text{slab}} - N\varepsilon_{\text{bulk}})/(2A)$. For both metals the stability order is $\gamma_{111} < \gamma_{001} < \gamma_{011}$, confirming that the close-packed (111) facet is the thermodynamically most stable surface. Ionic relaxation stabilises mainly the open (011) surface ($\Delta\gamma \approx -0.043 \text{ J m}^{-2}$ for Cu and -0.046 J m^{-2} for Pd), while the (111) facet is nearly unaffected and relaxation does not alter the stability ordering. The Wulff construction using the relaxed surface energies yields a truncated octahedron for Cu with $\{111\}$ (64.6%), $\{001\}$ (23.7%), and residual $\{011\}$ (11.7%) faces, and a near-perfect octahedron for Pd with $\{111\}$ (89.4%) and small $\{001\}$ (10.6%) faces only; the $\{011\}$ facet is entirely absent in Pd. The code used for the analysis is available in a public GitHub repository (see code section).

I. INTRODUCTION

The equilibrium shape of a crystalline nanoparticle is determined by the anisotropy of its surface free energy: each facet contributes an area proportional to its surface energy, and the shape that minimises the total surface free energy at fixed volume is given by the Wulff construction, first formulated in 1901. [1] For noble and transition metals crystallising in the fcc structure, the relevant low-index facets are (001) , (011) , and (111) , and their relative surface energies determine whether the equilibrium shape is dominated by octahedral $\{111\}$ faces, cubic $\{001\}$ faces, or a mixture of both. [2, 3]

Copper and palladium are paradigmatic fcc metals with very different catalytic applications: Cu is widely used for methanol synthesis and CO_2 reduction, while Pd is central to hydrogenation reactions and automotive exhaust catalysis. [7] Understanding the surface energetics of both metals at the atomistic level is therefore of practical as well as fundamental interest, since the equilibrium nanoparticle shape directly determines which surface facets — and hence which active sites — are exposed during a catalytic reaction.

In this report, DFT calculations within the generalised gradient approximation (GGA-PBE) are used to compute the surface energies of the three low-index facets for both Cu and Pd, and the Wulff construction is then applied to obtain the equilibrium nanoparticle shape. Symmetric, fully periodic slab models are employed throughout, and the effect of ionic relaxation is quantified by comparing fixed and relaxed surface energies.

II. EMPLOYED MODELS & METHODOLOGY

A. DFT setup

All calculations were performed with VASP 5.x using the PBE exchange–correlation functional [4] and projector-augmented wave (PAW) potentials. [5] The plane-wave kinetic-energy cutoff was set to 400 eV and the Brillouin zone was sampled with a Monkhorst–Pack k -point mesh adapted to each surface geometry. Structural relaxations were converged until the maximum force on any ion was below $0.01 \text{ eV \AA}^{-1}$.

B. Slab model

Each surface was modelled as a symmetric slab periodic in the in-plane directions and separated from its periodic images by a vacuum gap of at least $15 \text{ \AA}$. Symmetric slabs guarantee that both exposed surfaces are equivalent, which eliminates any artificial dipole perpendicular to the slab and allows the use of the standard surface energy expression

$$\gamma = \frac{E_{\text{slab}} - N\varepsilon_{\text{bulk}}}{2A}, \quad (1)$$

where E_{slab} is the total DFT energy of the slab, N is the number of atoms in the slab, $\varepsilon_{\text{bulk}}$ is the DFT bulk energy per atom, and A is the in-plane area of one surface. The factor of 2 accounts for the two equivalent surfaces of the symmetric slab. [6]

C. Fixed and relaxed surface energies

Two surface energies are computed for each facet. The *fixed* surface energy γ_{fix} is obtained from the first ionic

step energy, i.e. the energy of the slab with all atoms at their ideal bulk-terminated positions. The *relaxed* surface energy γ_{relax} uses the fully converged energy after ionic relaxation. The relaxation contribution is

$$\Delta\gamma = \gamma_{\text{relax}} - \gamma_{\text{fix}} \leq 0, \quad (2)$$

which is non-positive because relaxation can only lower the total energy. [3]

D. Wulff construction

The Wulff theorem states that the equilibrium shape of a crystal is the inner envelope of planes located at perpendicular distances from the centre proportional to the surface energy of each facet. [1] In practice, we define normalised h -values

$$h_{hkl} = \frac{\gamma_{hkl}}{\gamma_{\min}}, \quad (3)$$

where $\gamma_{\min} = \gamma_{111}$ is the lowest surface energy, so that $h_{111} = 1$. These h -values are entered into VESTA (*Edit* → *Edit Data* → *Crystal Shape*, space group Fm $\bar{3}$ m) to generate the equilibrium nanoparticle shape and read the absolute area of each face family from the VESTA output.

E. Computational details

The bulk reference energies were extracted from the OUTCAR of a conventional four-atom fcc unit cell: $\varepsilon_{\text{bulk}}^{\text{Cu}} = -3.725780 \text{ eV atom}^{-1}$ and $\varepsilon_{\text{bulk}}^{\text{Pd}} = -5.176240 \text{ eV atom}^{-1}$. Surface areas were computed from the lattice vectors in the POSCAR (or CONTCAR) of each slab, and the conversion $1 \text{ eV } \text{\AA}^{-2} = 16.0218 \text{ J m}^{-2}$ was used throughout. All post-processing was performed with the Python scripts `wulff_surface_energies.py` and `plots.py` available in the GitHub repository (see code section).

III. RESULTS AND DISCUSSION

A. Step 1 – surface energies of Cu and Pd

1. Fixed and relaxed surface energies

Table I lists the fixed and relaxed surface energies of all six slab systems studied in this work. For both Cu and Pd the stability ordering is

$$\gamma_{111} < \gamma_{001} < \gamma_{011}, \quad (4)$$

which is the expected sequence for fcc metals and is consistent with previously reported DFT-PBE values. [2, 3]

Figure 1 shows the same data as a grouped bar chart.

TABLE I: Fixed (γ_{fix}) and relaxed (γ_{relax}) surface energies and relaxation contribution ($\Delta\gamma$) for the low-index facets of Cu and Pd. All values in J m^{-2} . The most stable facet for each metal is marked with $\star$.

Metal	Surface	γ_{fix}	γ_{relax}	$\Delta\gamma$
Cu	(001)	1.5190	1.5102	−0.0088
Cu	(011)	1.5851	1.5421	−0.0430
Cu	(111)	1.3314	1.3313	−0.0001 $\star$
Pd	(001)	1.5044	1.5022	−0.0022
Pd	(011)	1.5871	1.5406	−0.0465
Pd	(111)	1.1333	1.1352	+0.0019 $\star$

figures/surface_energies_bar.pdf

FIG. 1: Fixed (γ_{fix} , light bars) and relaxed (γ_{relax} , dark bars) surface energies of Cu (blue) and Pd (orange) for the three low-index facets. The stability ordering $\gamma_{111} < \gamma_{001} < \gamma_{011}$ holds for both metals.

The (111) facet is the most stable surface for both metals. This is the close-packed surface of the fcc lattice, where each atom has nine nearest neighbours (six in-plane, three below), the highest coordination achievable at the surface. The small number of broken bonds per atom results in the lowest surface energy. By contrast, the (011) surface is the most open of the three, with rows of atoms separated by shallow troughs, which gives the highest surface energy.

The most notable quantitative difference between the two metals is the (111) surface energy: Pd(111) at 1.135 J m^{-2} is $\approx 0.20 \text{ J m}^{-2}$ lower than Cu(111) at 1.331 J m^{-2} . The (001) and (011) energies are very similar for both metals (within $\approx 0.05 \text{ J m}^{-2}$), consistent with the shared crystal structure and comparable d -band filling. [7]

2. Effect of ionic relaxation

Figure 2 shows the relaxation contribution $\Delta\gamma$ for each facet and metal.

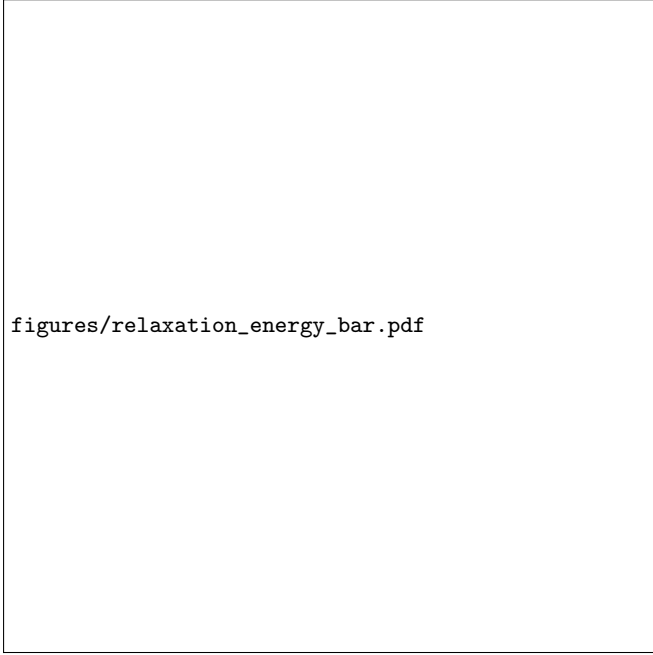


FIG. 2: Relaxation energy $\Delta\gamma = \gamma_{\text{relax}} - \gamma_{\text{fix}}$ (J m^{-2}) for Cu (blue) and Pd (orange). Negative values indicate relaxation-induced stabilisation. The (011) surface relaxes the most while (111) is nearly unaffected.

The relaxation effect is strongly facet-dependent. The open (011) surface relaxes the most: $\Delta\gamma = -0.043 \text{ J m}^{-2}$ for Cu and -0.046 J m^{-2} for Pd, because the under-coordinated atoms on this surface have more freedom to move inward and partially recover bulk-like coordination. By contrast, the close-packed (111) surface is nearly unaffected: $\Delta\gamma = -0.0001 \text{ J m}^{-2}$ for Cu and $+0.0019 \text{ J m}^{-2}$ for Pd. The marginal positive value for Pd (111) lies within the numerical noise of the DFT calculation and does not alter any physical conclusion. [6]

Crucially, ionic relaxation does not change the stability ordering of Eq. (4) for either metal. This answers the question directly: *relaxation is not a determining factor in the stability of the most stable surface*. The (111) facet is the most stable both before and after relaxation, because its close-packed geometry leaves virtually no room for the surface atoms to relax.

3. Wulff h -values

Table II lists the normalised Wulff h -values computed from the relaxed surface energies via Eq. (3). Figure 3 shows the same data as a bar chart.

For Pd, $h_{001} = 1.319$ and $h_{011} = 1.353$, both substantially larger than the Cu values of 1.134 and 1.158. This

TABLE II: Normalised Wulff h -values for Cu and Pd from the relaxed surface energies ($h_{hkl} = \gamma_{hkl}/\gamma_{111}$). These values are used as input for VESTA.

Metal	h_{001}	h_{011}	h_{111}
Cu	1.1344	1.1584	1.0000
Pd	1.3187	1.3527	1.0000

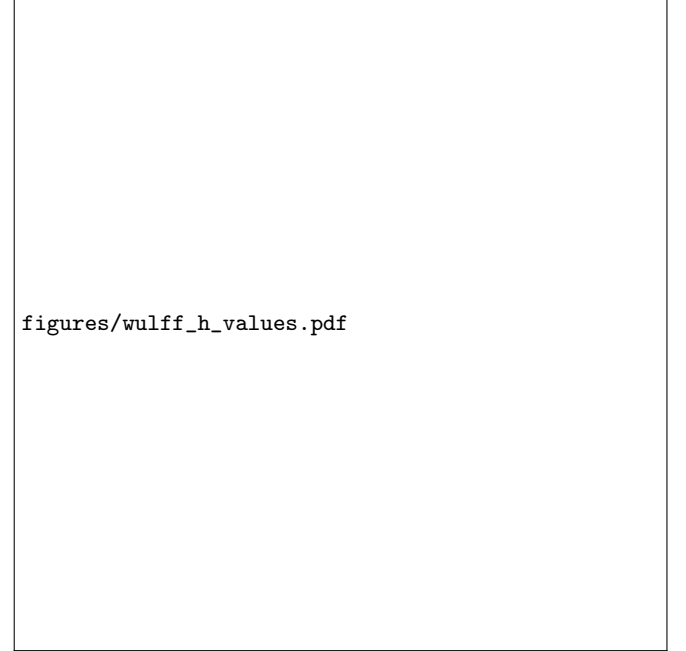


FIG. 3: Normalised Wulff h -values ($h_{111} = 1$) for Cu (blue) and Pd (orange). Larger h_{001} and h_{011} in Pd relative to Cu indicate that $\{111\}$ will dominate the Pd nanoparticle shape far more strongly.

already predicts that the non- $\{111\}$ faces will be proportionally smaller in Pd, leading to a shape closer to a regular octahedron.

B. Step 2 – Wulff shapes of Cu and Pd nanoparticles

1. Cu Wulff shape

Figure 4 shows the Wulff shape of a Cu nanoparticle obtained with the relaxed h -values from Table II.

The shape is a truncated octahedron, the classic equilibrium morphology of fcc metals. [2] Table III quantifies the exposed surface fractions from the VESTA face-area output. The $\{111\}$ faces account for 64.6% of the total surface area, the $\{001\}$ faces for 23.7%, and the $\{011\}$ faces for the remaining 11.7%. All three low-index facet families are exposed in the Cu equilibrium shape.

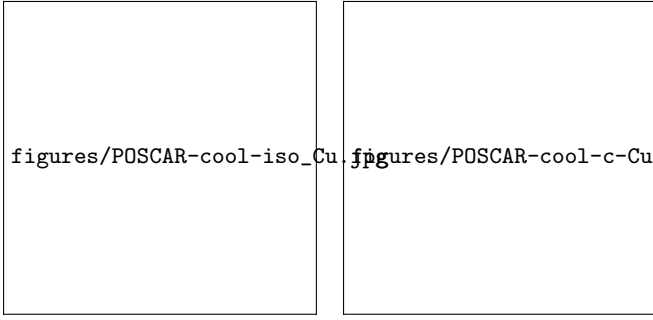


FIG. 4: Wulff shape of a Cu nanoparticle (VESTA, space group $Fm\bar{3}m$): isometric view (left) and view along $[001]$ (right). The shape is a truncated octahedron with large $\{111\}$ octahedral faces (dark grey) and smaller $\{001\}$ square faces (light grey), with residual $\{011\}$ facets.

2. Pd Wulff shape

Figure 5 shows the Wulff shape for Pd.

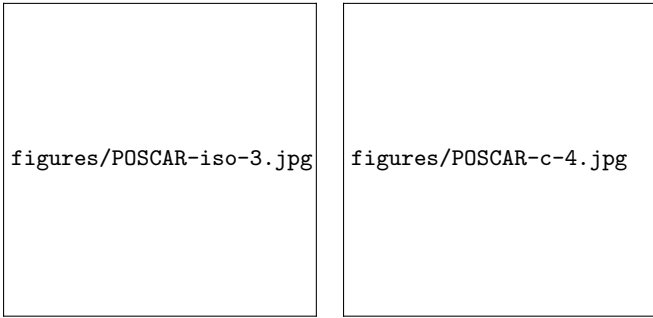


FIG. 5: Wulff shape of a Pd nanoparticle (VESTA, space group $Fm\bar{3}m$): isometric view (left) and view along $[001]$ (right). The shape is a near-perfect octahedron enclosed almost entirely by $\{111\}$ faces (pink) with small $\{001\}$ squares (orange). The $\{011\}$ facet is completely absent.

The Pd Wulff shape is strikingly different from Cu: it is a near-perfect truncated octahedron with $\{111\}$ faces dominating at 89.4% of the total area, and only small $\{001\}$ faces (10.6%) are present. Remarkably, the $\{011\}$ facet is completely absent. This occurs because the h_{011}/h_{111} ratio of 1.353 for Pd is large enough that the $\{011\}$ planes are pushed so far from the centre of the Wulff polyhedron that they are entirely cut out by the intersection of the $\{111\}$ and $\{001\}$ planes; the Wulff construction yields no $\{011\}$ face area at all.

TABLE III: Exposed surface area fractions of the Wulff equilibrium shapes of Cu and Pd, computed from the VESTA face-area output.

Metal	$\{111\}$ (%)	$\{001\}$ (%)	$\{011\}$ (%)
Cu	64.57	23.74	11.70
Pd	89.37	10.63	0.00

3. Role of relaxation in the Wulff shape

To assess whether ionic relaxation plays a determining role in the nanoparticle morphology, we compare the fixed and relaxed h -values. For Cu, using γ_{fix} instead of γ_{relax} gives $h_{001}^{\text{fix}} = 1.141$ vs. $h_{001}^{\text{relax}} = 1.134$ and $h_{011}^{\text{fix}} = 1.191$ vs. $h_{011}^{\text{relax}} = 1.158$. For Pd the differences are similarly small. These shifts correspond to changes of at most 3% in the h -values and do not qualitatively change either Wulff shape.

Relaxation is therefore *not* a determining factor in the Wulff shape. The shapes obtained from fixed and relaxed surface energies are qualitatively identical: both give a truncated octahedron for Cu and a near-octahedron for Pd, with only minor differences in the relative sizes of the $\{001\}$ faces. The dominant effect is the large intrinsic anisotropy between γ_{111} and $\gamma_{001}/\gamma_{011}$, which is already well captured by the unrelaxed calculation.

4. Implications for catalysis

The Wulff construction directly determines which surfaces are available for catalytic reactions. For Cu nanoparticles, all three facet families — $\{111\}$, $\{001\}$, and $\{011\}$ — are present on the equilibrium shape, so all three should be considered when studying the catalytic activity of Cu nanoparticles. The $\{111\}$ surface dominates (64.6%) but the $\{001\}$ contribution (23.7%) is large enough to be catalytically significant, and even the small $\{011\}$ strips (11.7%) may play a role in structure-sensitive reactions. [7]

For Pd nanoparticles the situation is fundamentally different. Only $\{111\}$ (89.4%) and $\{001\}$ (10.6%) are exposed; the $\{011\}$ surface is entirely absent. In practice this means that catalytic studies on Pd nanoparticles need only consider $\{111\}$ and $\{001\}$ active sites, unless the nanoparticle is far from its equilibrium shape (e.g. under reaction conditions or synthesised by a kinetically controlled route). The very strong $\{111\}$ dominance of Pd also means that the catalytic activity of Pd nanoparticles will be especially sensitive to any changes in the $\{111\}$ surface energy, for example due to adsorbate coverage or alloying. [7]

IV. CONCLUSIONS

DFT-PBE calculations on symmetric fcc slabs give the stability ordering $\gamma_{111} < \gamma_{001} < \gamma_{011}$ for both Cu and Pd, with the (111) surface being the most stable for each metal. Ionic relaxation is strongest for the open (011) surface ($\Delta\gamma \approx -0.04 \text{ J m}^{-2}$) and negligible for (111) , but does not alter the ordering and is therefore not a determining factor in either the surface stability or the Wulff shape.

The Wulff constructions reveal two qualitatively distinct nanoparticle morphologies. Cu adopts a truncated

octahedron with all three facet families present ($\{111\}$ 64.6%, $\{001\}$ 23.7%, $\{011\}$ 11.7%). Pd adopts a near-perfect octahedron dominated by $\{111\}$ (89.4%) with only small $\{001\}$ faces (10.6%) and no $\{011\}$ at all. This striking difference follows directly from the much larger anisotropy of the Pd surface energies: the $\gamma_{011}/\gamma_{111}$ ratio of 1.353 for Pd is sufficient to eliminate the $\{011\}$ face entirely from the Wulff polyhedron.

These results have a direct implication for catalysis: studies of Cu nanoparticles must account for three facet

types, while those of Pd need consider only $\{111\}$ and $\{001\}$. The stronger $\{111\}$ preference of Pd is consistent with the widely observed tendency of Pd nanoparticles to adopt octahedral shapes in synthesis experiments.

Code: GitHub repository

<https://github.com/itxasoma/V2--Wulff-Shapes>

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- [1] G. Wulff, “Zur Frage der Geschwindigkeit des Wachstums und der Auflösung der Krystallflagen,” *Z. Kristallogr.* **34**, 449–530 (1901).
 - [2] L. Vitos, A. V. Ruban, H. L. Skriver, and J. Kollár, “The surface energy of metals,” *Surf. Sci.* **411**, 186–202 (1998).
 - [3] M. Methfessel, D. Hennig, and M. Scheffler, “Trends of the surface relaxations, surface energies, and work functions of the 4d transition metals,” *Phys. Rev. B* **46**, 4816–4829 (1992).
 - [4] J. P. Perdew, K. Burke, and M. Ernzerhof, “Generalized gradient approximation made simple,” *Phys. Rev. Lett.* **77**, 3865–3868 (1996).
 - [5] P. E. Blöchl, “Projector augmented-wave method,” *Phys. Rev. B* **50**, 17953–17979 (1994).
 - [6] V. Fiorentini and M. Methfessel, “Extracting crystal surface energies from *ab initio* calculations,” *J. Phys.: Condens. Matter* **8**, 6525–6529 (1996).
 - [7] J. K. Nørskov, T. Bligaard, J. Rossmeisl, and C. H. Christensen, “Towards the computational design of solid catalysts,” *Nat. Chem.* **1**, 37–46 (2009).